

CPN4M: Testing Multi-Agent Systems under Organizational Model Moise+ Using Colored Petri Nets

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1. RESEARCH OBJECTIVE & SYSTEM ARCHITECTURE

Colored Petri Net (CPN) model checking tool to generate test cases verifying role compliance, obligations, and permissions under Moise+ organizational specifications.

2. BENCHMARK DATASETS & EMPIRICAL EVALUATION

JaCaMo multi-agent implementation of collaborative 'Writing Paper' benchmark.

Key Numerical Findings & Empirical Proofs:

Accurately detects obligation/permission deadlocks and role non-compliance during multi-agent collaboration.

3. CRITICAL ARCHITECTURAL FLAWS & METHODOLOGICAL LIMITATIONS

Identified System Failure Modes:

Rigid static XML organizational specifications; state-space verification explodes exponentially on more than 5 agents; no adaptive role bypassing.

4. RESEARCH GAP & RELEVANCE TO CAG-DELPHI

This work exemplifies the *Unconditional Consultation Dilemma* observed across the 15-paper corpus. By failing to incorporate epistemic uncertainty self-gating prior to multi-agent activation, the architecture incurs severe token inflation and exposes confident primary solutions to debate degeneration noise. In CAG-Delphi, this paper serves as an empirical anchor for diversity-governed consensus and comparative baseline evaluation.

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